

[14X030] Introduction to Computational Finance

Exercise # 8

April 21, 2026

General instructions

Each student is expected to upload on Moodle a **zip** file named as *firstname_lastname*, containing the followings:

- A report in **pdf** format to answer exercise questions.
- The **code** used to generate the results of the report.
- If preferred, submit a unique **Jupyter notebook** with both code and report explanations.

Deeper instructions

- **Report:**
 - Answers to the TP questions. Figures and numerical results are necessary but **strictly not** sufficient: provide your analysis/comments as well.
 - Example: if your results surprise you, explain why and explain what the expected result was.
 - **Pay attention** to the presentation: axis labels, legend, etc.
- **Code:**
 - Implementation questions + code used to produce the results and figures in the report.
 - Programming language of your choice.
 - Do not use ChatGPT, or any other LLM/AI-assisted tools while writing down your code. Try by yourself, otherwise it will be counted as failed.
 - I **strongly recommend Python/Julia**: I won't be able to assist you in the same way if you use a different language.

Deadline

Any questions to: [Lorenzo Bini](#).

Upload on Moodle due by : **April 27, 2026 at 11:59 pm**

Minority Game

- Let N be the number of agents, M the number of bits of history and S the number of strategies available to each agent among the 2^{2^M} possible strategies.
- Each agent initializes the utilities of its strategies to zero.
- Initialize the history $\mu(0)$ to a random list of M bits.
- For t in $1, \dots, T$:
 - Each agent $i \in \{1, \dots, N\}$ samples a strategy $s_i(t)$ according to the softmax distribution of utilities: $\frac{\exp(\Gamma_i u_{s_i}(t))}{\sum_{s'} \exp(\Gamma_i u_{s'}(t))}$ where $\Gamma_i > 0$.
 - Given the current history $\mu(t)$, each agent uses its chosen strategy $s_i(t)$ to pick an action $a_i(t) \in \{+1, -1\}$.
 - Compute the attendance $A(t) = \sum_{i=1}^N a_i(t)$.
 - Update the utility of the chosen strategies with a linear payoff:

$$u_{s_i(t)} = u_{s_i(t-1)} - a_i(t) \frac{A(t)}{\beta}$$

- Remove the oldest bit of history and add a new one.
1. (a) Why is the above procedure called a minority game?
 (b) What is the role of Γ_i ? In particular, what does a large or a small value of Γ_i means?
 2. Implement a minority game. You can use $\beta = 1$ and $\Gamma_i = 0.01, \forall i \in \{1, \dots, N\}$. To add a new bit of history, you can either pick it at random or from some function of the attendance (1 if positive attendance, 0 if negative attendance for instance).
 3. Simulate a minority game with $S = 2$ strategies for $T = 100$ steps for values of N in $\{51, 101, 251, 501, 1001\}$ and values of M in $\{0, 1, \dots, 18\}$. On a log-log plot, represent $\frac{\sigma^2}{N}$, the scaled variance of the attendance, against $\alpha = \frac{2^M}{N}$.
 4. What is the critical value α_c for which the volatility reaches a minimum?
 5. Optional: Define an initial price $p(0)$ (for instance 100) and then update it as follows: $p(t) = p(t-1) \exp(\frac{A(t)}{\lambda})$ with λ some positive constant. What is the intuition behind this update rule? Plot price curves for different values of α and λ and comment.